1st Place: Aspect 7 – Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations

- **Analysis**: This is the most central operational skill for this problem. Students need to directly manipulate a complex geometric figure (the shop floor plan), which includes an internal region (service area and counter) that needs to be excluded. They must use the geometric representation (grid) to obtain area information.
- **Manifestation in the Question**: Students need to view the floor plan as a composite figure, potentially decomposing it into simpler shapes (e.g., rectangles, triangles) or employing a "whole minus part" strategy. They must process grid data to obtain dimensions of various parts and perform corresponding area calculations.

2nd Place: Aspect 12 – Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

- **Analysis**: Calculating the net area involves a clear, ordered multi-step logical process.
- **Manifestation in the Question**: A typical problem-solving procedure is:
  1. **Identify the Whole**: Determine the **overall outline** of the shop (including the service area) and calculate its total area (`A_total`).
  2. **Identify the Part**: Determine the outline of the **service area (including counter)** and calculate its area (`A_service`).
  3. **Perform the Operation**: Calculate the net area: `A_floor = A_total - A_service`.
  4. **Measure/Calculate**: Each of the above steps may require further sub-steps, such as decomposing figures, counting squares, applying area formulas, and unit conversion.
  5. **State**: Provide the final area with units.

3rd Place: Aspect 4 – Make mathematical diagrams, graphs, constructions or computing artifacts and extracting mathematical information from them

- **Analysis**: To clarify their thinking, students likely need to actively manipulate the graphic on the original diagram or on scratch paper.
- **Manifestation in the Question**: Students may need to **outline or label** the overall contour and the service area contour differently on the diagram, or **partition/complete** irregular overall or service area shapes into regular shapes (e.g., rectangles) to facilitate applying area formulas. This is "making" or reconstructing a mathematical diagram that is easier to analyze and extracting key length and area information from it.

4th Place: Aspect 8 – Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

- **Analysis**: The requirement to "Show your work" means students must clearly present the entire process from identifying shapes to the final calculation.
- **Manifestation in the Question**: In their solution, students should show intermediate results, such as: the process for calculating the dimensions and area of the overall region, the process for calculating the dimensions and area of the service region, and the final subtraction formula. This demonstrates clear problem-solving logic and a logical chain.

5th Place: Aspect 6 – Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms

- **Analysis**: The core of solving the problem lies in applying geometric concepts of area and related formulas.
- **Manifestation in the Question**: Students need to accurately apply the definition of **area** (the size of a plane figure). They need to apply basic area formulas, such as **rectangle area = length × width**. Because a grid is used (each square is 0.5m × 0.5m, i.e., 0.25 m^2), they also need to understand the rules for **area unit conversion**. All this relies on mastery of the mathematical formal system.

6th Place: Aspect 10 – Make sense of, relate and use a variety of representations when interacting with a problem

- **Analysis**: The problem integrates visual graphical representation (gridded floor plan), symbolic/numerical representation (area formulas, calculation process), and textual representation (problem description).
- **Manifestation in the Question**: Students must understand the textual instruction ("excluding the service area and counter"), identify the corresponding regions on the visual graphic, and translate these regions into numerical dimensions and area values usable for calculation. They need to establish accurate correspondences between these representations.

7th Place: Aspect 13 – Use an understanding of the context to guide or expedite the mathematical solving process, e.g., working to a context-appropriate level of accuracy

- **Analysis**: The real-world context of "putting new flooring" directly influences the problem-solving strategy and precision.
- **Manifestation in the Question**: The context helps students understand that "total floor space area" refers to the **net area of usable floor**, not the built-up area or gross area. It may also prompt students to consider practical calculation: since it's for estimating materials, the result might require a reasonable approximation (e.g., rounded to one decimal place or to the nearest 0.25 m^2). Understanding "excluding the service area" implies using a subtraction strategy.

8th Place: Aspect 11 – Switch between different representations in the process of finding solutions

- **Analysis**: The problem-solving process embodies dynamic switching between different representations.
- **Manifestation in the Question**: Switching from **textual instructions** to **graphical identification** (circling areas on the diagram); from **graphical information** (number of squares) to **numerical information** (lengths, areas); from **partial numerical information** (areas of parts) via **arithmetic operations** (subtraction) to the **final numerical answer**. This chain of conversions is key to solving the problem.

第1位：方面 7 - 处理数字、图形与几何表示

- **分析**：这是本题最核心的操作能力。学生需要直接处理一个复杂的几何图形（店铺平面图），该图形包含一个需要被排除的内部区域（服务区与柜台）。他们必须运用几何表示（网格）来获取面积信息。
- **题目体现**：学生需要将平面图视为一个组合图形，可能将其分解为多个简单图形（如矩形、三角形），或采用“整体减去部分”的策略。他们必须处理网格数据来获取各部分的尺寸，并进行相应的面积计算。

第2位：方面 12 - 使用多步骤流程得出数学解答或结论

- **分析**：计算净面积涉及一个明确、有序的多步逻辑过程。
- **题目体现**：一个典型的解题步骤是：
  1. **识别整体**：确定店铺（包括服务区）的**整体轮廓**，并计算其总面积（`A_total`）。
  2. **识别部分**：确定**服务区（含柜台）** 的轮廓，并计算其面积（`A_service`）。
  3. **执行运算**：计算净面积：`A_floor = A_total - A_service`。
  4. **测量/计算**：上述每一步都可能需要进一步的子步骤，如分解图形、数格子、应用面积公式、单位换算。
  5. **表述**：得出最终面积并给出单位。

第3位：方面 4 - 制作数学图表、图形或构造物，并从中提取数学信息

- **分析**：为了理清思路，学生很可能需要在原图或草稿上进行主动的图形操作。
- **题目体现**：学生可能需要**在图上用不同方式勾勒或标注**整体轮廓和服务区轮廓，或**将不规则的整体或服务区图形分割/补全成规则图形**（如矩形），以便于应用面积公式。这就是在“制作”或重构一个更易分析的数学图形，并从中提取关键的长度和面积信息。

第4位：方面 8 - 阐明解决方案，展示和/或总结、呈现中间数学结果

- **分析**：“Show your work”的要求意味着学生必须清晰地展示从识别图形到最终计算的全过程。
- **题目体现**：在解答中，学生应展示中间结果，例如：整体区域的尺寸和面积计算过程、服务区域的尺寸和面积计算过程、以及最终的减法算式。这体现了清晰的解题思路和逻辑链条。

第5位：方面 6 - 理解并运用基于定义、规则和形式系统的构造

- **分析**：解决问题的核心是运用几何中的面积概念和相关公式。
- **题目体现**：学生需要准确运用**面积**的定义（平面图形的大小）。他们需要应用基本的面积公式，如**矩形面积 = 长 × 宽**。由于使用了网格（每格0.5米×0.5米，即0.25平方米），他们还需要理解**面积单位的换算**规则。这都依赖于对数学形式系统的掌握。

第6位：方面 10 - 在解决问题时理解、关联和使用多种表征

- **分析**：题目融合了视觉图形表征（网格平面图）、符号/数字表征（面积公式、计算过程）和文本表征（问题描述）。
- **题目体现**：学生必须理解文本指令（“excluding the service area and counter”），在视觉图形上识别出相应的区域，并将这些区域转化为可用于计算的数字尺寸和面积值。他们需要在这些表征之间建立准确的对应关系。

第7位：方面 13 - 利用对情境的理解来指导或加速求解过程

- **分析**：“铺新地板”这一现实情境直接影响解题策略和精度。
- **题目体现**：情境帮助学生理解“total floor space area”指的是**可用地面的净面积**，而非建筑面积或毛面积。它也可能提示学生考虑计算的实用性：由于是估算材料，结果可能需要合理的近似（例如，保留到小数点后一位或取整到最接近的0.25平方米）。理解“排除服务区”意味着使用减法策略。

第8位：方面 11 - 在求解过程中在不同表征间转换

- **分析**：解题过程是不同表征间动态转换的体现。
- **题目体现**：从**文本指令**转换到**图形识别**（在图上圈出区域）；从**图形信息**（格子数）转换到**数字信息**（长度、面积）；从**部分数字信息**（各部分面积）通过**算术运算**（减法）转换到**最终的数字答案**。这个转换链条是解题的关键。